



# TECHNICAL REPORT

Team# 6

ECLIFT

01/05/2026



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LYON**

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## ABSTRACT

For the XtraChallenge 2026 competition, the student team ECLift from École Centrale Lyon developed a fixed-wing RC aircraft designed to meet mission requirements of payload transport and aerodynamic efficiency. The project involved a structured approach including subsystem selection, aerodynamic design using XFLR5, stability analysis, structural design, and overall performance evaluation. The aerodynamic configuration was optimized to achieve a balance between lift generation, drag reduction, and stability. Static and dynamic stability analyses confirmed predictable and stable flight behavior, while performance calculations demonstrated adequate cruise speed, endurance, and payload capability. Overall, the aircraft design meets the competition objectives and represents a well-balanced and feasible solution for the intended mission.

## Nomenclature

### Aerodynamic Coefficients

|             |                             |
|-------------|-----------------------------|
| $C_L$       | Lift coefficient            |
| $C_D$       | Drag coefficient            |
| $C_m$       | Pitching moment coefficient |
| $C_{L,max}$ | Maximum lift coefficient    |

### Flight Variables

|           |                 |
|-----------|-----------------|
| $V$       | Velocity        |
| $V_s$     | Stall speed     |
| $V_{TO}$  | Take-off speed  |
| $V_{max}$ | Maximum speed   |
| $\alpha$  | Angle of attack |

### Forces and Performance

|         |                        |
|---------|------------------------|
| $T$     | Thrust                 |
| $W$     | Weight                 |
| $T/W$   | Thrust-to-weight ratio |
| $ROC$   | Rate of climb          |
| $Range$ | Flight range           |

### Geometry

|       |                        |
|-------|------------------------|
| $b$   | Wingspan               |
| $S$   | Wing area              |
| $c$   | Chord length           |
| $MAC$ | Mean aerodynamic chord |

### Propulsion

|          |                  |
|----------|------------------|
| $P$      | Power            |
| $\delta$ | Throttle setting |
| $J$      | Advance ratio    |

### Constants

|        |             |
|--------|-------------|
| $\rho$ | Air density |
|--------|-------------|

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# 1 Introduction

ECLift is the aerospace club at Centrale Lyon focused on the design and development of radio-controlled aircraft and drones. It brings together students interested in aerodynamics, structures, and flight systems, providing a hands-on platform to apply theoretical knowledge through practical projects and competitions.

For the XtraChallenge 2026 competition, the team presents **ECLifter 26**, a fixed-wing electric aircraft designed to meet the requirements of both efficient and urgent transport missions. The objective is to achieve a balance between payload capacity, aerodynamic efficiency, and reliability while respecting the constraints imposed by the competition.

The design follows a simple and efficient approach, with emphasis on minimizing drag, ensuring stable flight under varying payload conditions, and maintaining ease of assembly and operation. This report presents the development of the aircraft and the reasoning behind the key design choices.

## 1.1 Team Organization

The ECLift team consists of 10 students from Centrale Lyon, including 8 from the Master's program in Aerospace Engineering and 2 from the Ingénieur Généraliste program.

The team was divided into three main divisions: Aerodynamics, Structures, and Materials & Manufacturing. Each group focused on its respective area, while regular coordination ensured consistency across the overall aircraft design.

| AERODYNAMICS TEAM                 |                                            |               |
|-----------------------------------|--------------------------------------------|---------------|
| Shawn Varghese Sunil<br>TEAM LEAD | Pushkar Kolhar<br>ELECTRONICS & PROPULSION | Claudia       |
| STRUCTURES TEAM                   |                                            |               |
| Marie Castéran<br>STRUCTURES LEAD | Zynab Osman                                | Mateo Jund    |
| Paavan Miteshkumar Kalyani        |                                            |               |
| MATERIALS & FABRICATION TEAM      |                                            |               |
| Karim Eltannikhy                  | Carlos Macías Toril                        | Hugo Djelloul |

Figure 1: Organization of the team

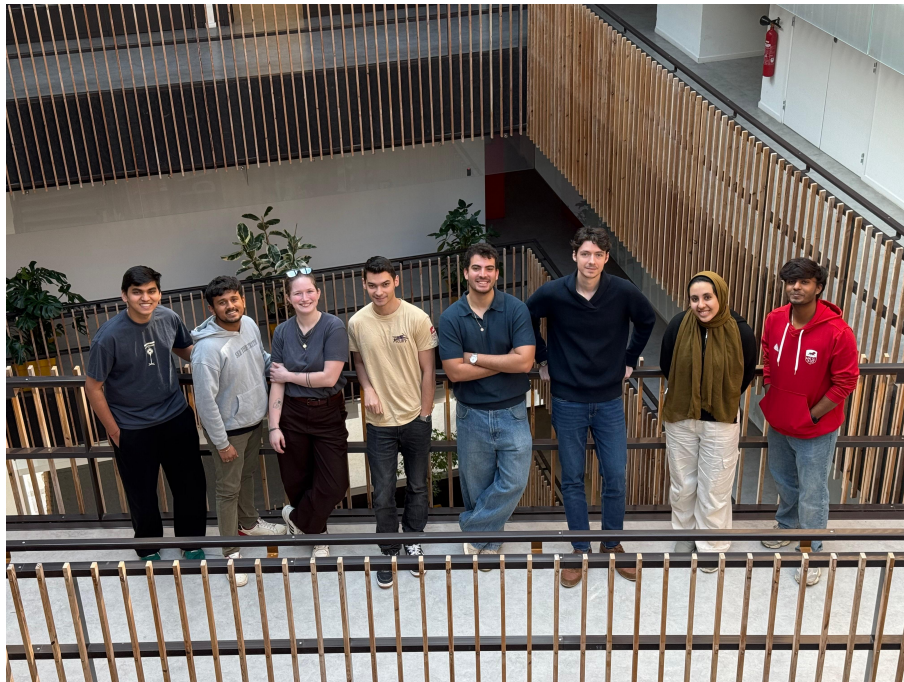


Figure 2: Team members of ECLIFT

## 2 Project Management

A detailed chart was developed to clearly define the project timeline, key milestones, and deadlines. It outlines the progression of the project from initial concept development and preliminary design phases to detailed design, manufacturing, and testing. This planning tool helped the team organize tasks efficiently, track progress, and ensure that all stages of the project were completed within the required deadlines.

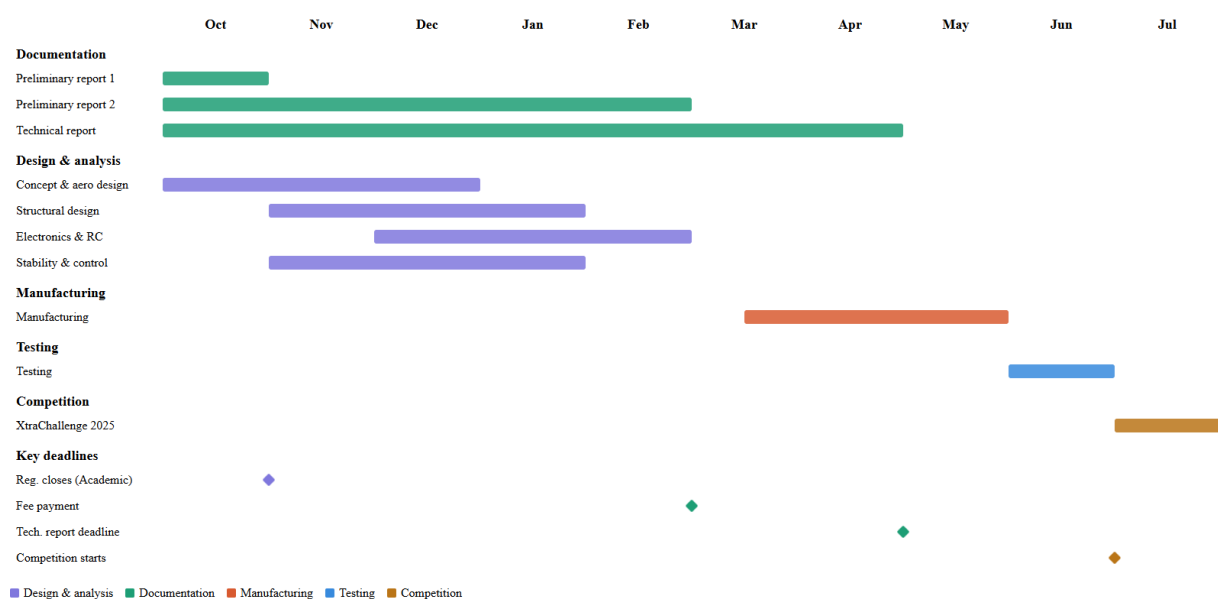


Figure 3: Gantt Chart

## 2.1 Project Expenses

The aircraft budget is divided into two main categories: electronics and propulsion, and materials and fabrication. The objective of the budget optimization was to reduce the total cost to approximately 400 EUR while maintaining flight safety, structural integrity, and overall performance.

### 2.1.1 Total Aircraft Cost

Table 1: Total estimated aircraft cost

| Category                      | Cost [EUR] |
|-------------------------------|------------|
| Electronics and propulsion    | 247        |
| Materials and fabrication     | 156        |
| <b>Total estimated budget</b> | <b>403</b> |

## 3 Selection of Subsystems

### 3.1 Propeller Selection

During the propeller selection, we considered both diameter and pitch simultaneously. We evaluated six candidate configurations against two governing constraints. The Thrust to weight ratio should be  $T/W \geq 1.0$ , and a current draw below 48 A, which is a safe operating condition for the 60 A ESC.

For each configuration, we analyzed the electrical power scales with multiple diameters and pitches based on the blade element theory:

$$P \propto D^4 \cdot p \quad (1)$$

Static thrust is also estimated for different propeller using actuator disk theory. We fixed the parameters of the motor and pre-analyzed battery configuration and calculated thrust:

$$T = C \cdot (2\rho A P_{\text{shaft}}^2)^{1/3} \quad (2)$$

The advance ratio  $J$  is used as an indicator of cruise efficiency. In simple words,  $J$  is used to determine how hard the propeller is working to pull the aircraft forward. Peak propeller efficiency occurs in the range  $J = 0.6\text{--}0.8$ .

$$J = \frac{V_{\text{cruise}}}{V_{\text{pitch}}}, \quad V_{\text{pitch}} = \frac{p \cdot \text{RPM}}{1056} \quad (3)$$

| Propeller | P[W] | I[A] | T[kgf] | T/W   | J     | motor eff[%] | prop eff[%] |
|-----------|------|------|--------|-------|-------|--------------|-------------|
| 8×5       | 270  | 18.8 | 1.735  | 0.526 | 0.771 | 91.1         | 40.6        |
| 10×4      | 528  | 36.7 | 2.949  | 0.893 | 0.963 | 82.7         | 6.2         |
| 10×5      | 660  | 45.8 | 3.301  | 1     | 0.771 | 78.4         | 36.8        |
| 10×6      | 792  | 55   | 3.589  | 1.088 | 0.642 | 74           | 55          |
| 11×5      | 966  | 67.1 | 4.139  | 1.254 | 0.771 | 68.3         | 36.1        |
| 12×5      | 1369 | 95   | 4.794  | 1.453 | 0.771 | 55.1         | 36.6        |

Figure 4: Propeller configuration comparison

Smaller diameters and lower pitches consistently fall short of the minimum thrust required for takeoff. In the case of larger diameters and higher pitches, they generate adequate thrust but overload the ESC beyond its safe operating point. The 10×5 is the only configuration that sits at the intersection of both constraints. That is producing enough thrust while keeping the current within safe limits. Its advance ratio is of 0.771 which confirms that it operates in the peak efficiency region during cruise. This makes it the optimal choice for the mission.

### 3.2 Thrust Model

The thrust is modelled as a function of both airspeed and throttle setting. At a given throttle  $\delta \in [0, 1]$ , the motor rotational speed scales linearly, and since propeller thrust varies with the square of rotational speed, the static thrust is:

$$T(V, \delta) = T_{\max} \cdot \delta \cdot (\delta - k V) \quad (4)$$

The model works on the assumptions that resistive losses in the motor and ESC are treated as constants. The blade stalls are neglected.

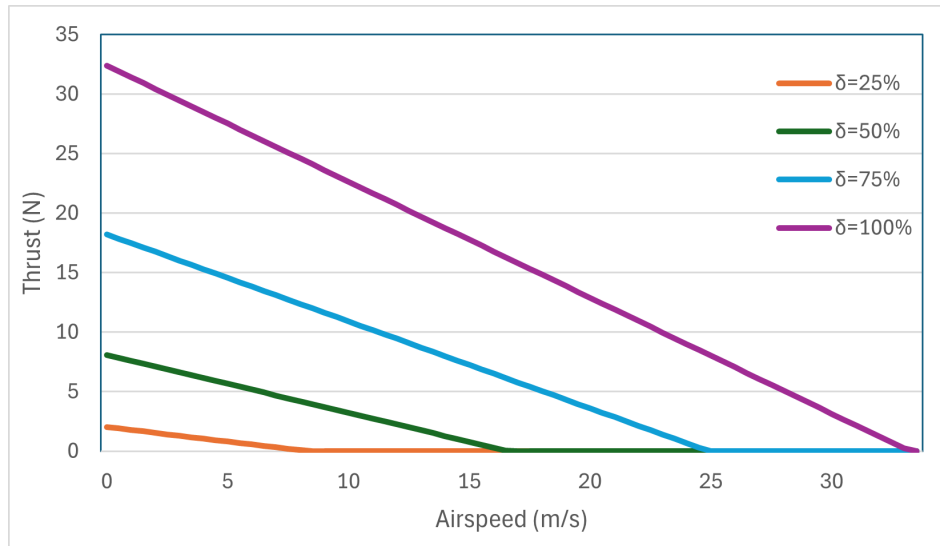


Figure 5: Thrust as a function of airspeed for four throttle settings.

The resulting comparison of thrust curves is shown in Figure 5. Each line corresponds to a fixed throttle level. At full throttle, the aircraft operates at a cruise speed of approximately 25.6 m/s. The model also predicts that a thrust of 7.4 N is produced at the same cruise speed. This can be considered sufficient to maintain level flight at the design weight. But to avoid heating and power loss, we will be flying at low throttle level. This can result to an approximate air speed of 10-12 m/s at cruise level, which can very depending on the terrains and piloting capabilities.

### 3.3 Avionics Architecture

The avionics architecture of the aircraft is illustrated in Figure 6. The system is organised around two fully independent power rails to ensure reliable operation of all subsystems throughout the flight.

The complete list of selected avionics components is as follows:

- Tattu LiPo 4S 2300 mAh — Main battery

- Tattu LiPo 2S 650mAh HV — Secondary battery
- Skywalker 60A V2 — Electronic Speed Controller (ESC)
- Emax GT2826/04 1090 KV — Brushless motor
- EMAX Servo  $\times 4$  — Control surface actuators
- UBEC 5A — Voltage regulator
- FlySky FS-i6 — Radio transmitter
- FlySky FS-IA6B — Receiver
- Telemetry Box — Onboard monitoring system
- 40 A Safety Switch

### 3.3.1 Power Architecture.

The main battery drives the propulsion chain only and connects directly to the Skywalker ESC, which drives the brushless motor. During thrust testing, a 40A ESC burned out despite the current not exceeding 40A. As a result, the ESC was upgraded to a 60A version for further testing and operation. All control electronics are powered independently by the Auxiliary battery, which passes through a 40 A safety switch. To this a 5A UBEC is connected to deliver stable power supply to the onboard electronics. This dual architecture guarantees that, in case of a fault on the main propulsion battery, the power supply has no effect on the control surfaces. This ensures that the pilot has full control of the aircraft at all times.

### 3.3.2 Control and Telemetry.

The FlySky FS-i6 transmitter communicates with the FS-IA6B receiver over 2.4 GHz band. The receiver delivers commands to the four servos which controls the aircraft. The ESC also receives its throttle signal from the receiver. Telemetry box collects onboard flight data and sends it back to the on-ground receiver.

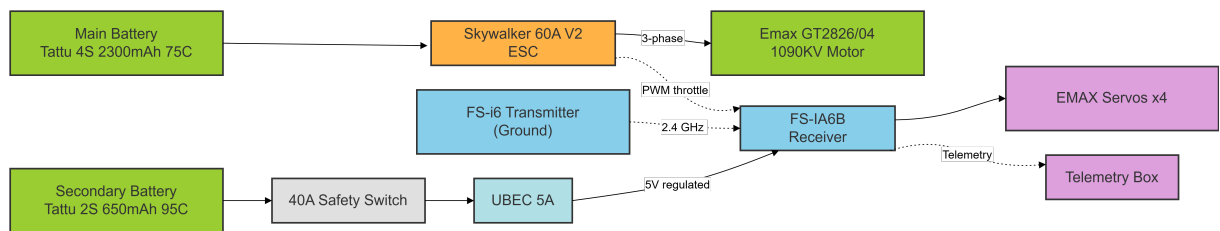


Figure 6: Avionics architecture.

In the figure 6, Solid lines indicates power connections whereas the dashed lines indicate signal paths. The architecture carefully prioritizes the signal and power supply chains to avoid power wastage and proper safety.

## 3.4 Battery Sizing

### 3.4.1 Propulsion System Characteristics

The propulsion system calculations are driven by the Emax GT2826/04 1090 KV brushless motor. The analytical powertrain model developed in Section 2.2 provides the current draw across all

mission phases. Which gives us peak takeoff current of 24.35 A at full throttle. Since this current is drawn directly by the motor through the ESC, it governs the specification of the battery components.

### 3.4.2 Main Battery

The main battery is a Tattu LiPo 4S 2300 mAh 75C. To verify if it provides enough power, the flight mission is divided into three phases:

$$Q = I \cdot \frac{t}{3600} \quad (5)$$

| Phase        | Duration[s] | Current[A] | Charge[mAh]   |
|--------------|-------------|------------|---------------|
| Takeoff      | 60          | 24.35      | 405.8         |
| Cruise       | 150         | 14.90      | 620.8         |
| Landing      | 40          | 4.87       | 54.1          |
| <b>Total</b> | <b>250</b>  | —          | <b>1080.7</b> |

Table 2: Mission profile and energy consumption per flight phase.

The 75C discharge rating allows a maximum continuous current of:

$$I_{\max} = 2.3 \times 75 = 172.5 \text{ A} \gg 24.35 \text{ A} \quad (6)$$

confirming that the battery has a large thermal buffer for high thrust maneuvers. For safety, if we apply an 80% depth-of-discharge limit, the usable capacity becomes:

$$C_{\text{usable}} = 2300 \times 0.80 = 1840 \text{ mAh} \quad (7)$$

The remaining reserve after the mission is:

$$C_{\text{reserve}} = 1840 - 1080.7 = 759.3 \text{ mAh} \quad (8)$$

This provides a comfortable safety margin for high load and long flight possibilities.

### 3.4.3 Secondary Battery

The electronics line is powered by a Tattu LiPo 2S 650 mAh 95C battery. This secondary battery has to go through a 40 A safety fuse switch before reaching the electronics. It provides an option of manual cutoff and protection to the avionics circuit. Connected to the fuse, there is a 5A UBEC. It regulates voltage and distributes power to the receiver, telemetry box, and four EMAX servos.

Each servo draws approximately 0.18 A under typical flight load, giving:

$$I_{\text{servos}} = 4 \times 0.18 = 0.72 \text{ A} \quad (9)$$

Even during high power maneuvers with all four servos active, the electronics load remains below 1.5 A. This is well within the capability of both the 5 A UBEC and the secondary battery. This is the advantage of having an isolated power rail architecture. It ensures that control is maintained regardless of the state of the main propulsion battery.

## 4 Aerodynamic Design

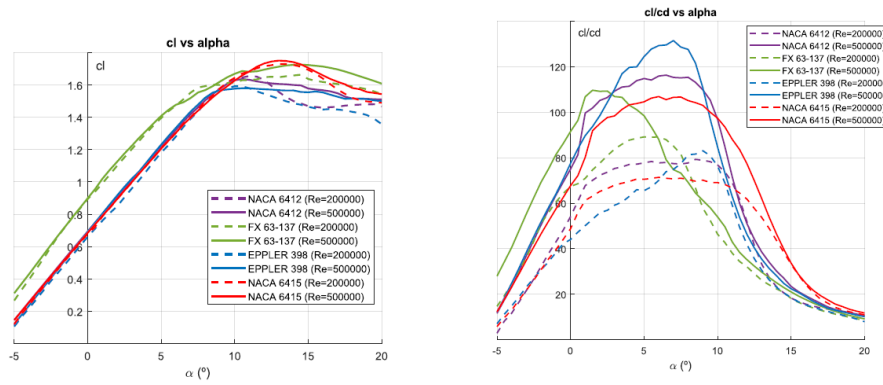
The aerodynamic configuration of the aircraft was developed through an iterative design process, combining analytical methods and numerical simulations using XFLR5. The primary objective was to achieve a balance between high lift capability for payload missions and aerodynamic efficiency for endurance performance. The design process involved the selection and optimization of the airfoil, wing geometry, winglets, and empennage configuration, ensuring stable and efficient flight across the operational envelope.

### 4.1 Airfoil Design

The aerodynamic performance of the aircraft is strongly influenced by the selection of the airfoil, particularly under low Reynolds number conditions typical of RC-scale flight. Therefore, a systematic evaluation of candidate airfoils was conducted using XFLR5, where lift coefficient ( $C_L$ ), drag coefficient ( $C_D$ ), and pitching moment coefficient ( $C_M$ ) were analyzed over a range of angles of attack.

The analysis was performed at Reynolds numbers representative of the expected flight regime ( $Re \approx 2 \times 10^5$ ). Airfoils with thickness ranging between 12% and 15% and moderate camber were considered to ensure sufficient structural thickness while maintaining aerodynamic efficiency.

Several airfoils, including NACA 6412, NACA 6415, Eppler 66, Eppler 210, and Eppler 398, were evaluated based on maximum lift coefficient ( $C_{L_{max}}$ ), lift-to-drag ratio ( $L/D$ ), drag characteristics, and stability behavior.



(a)  $C_L$  vs  $\alpha$

(b) Eficiencia vs  $\alpha$

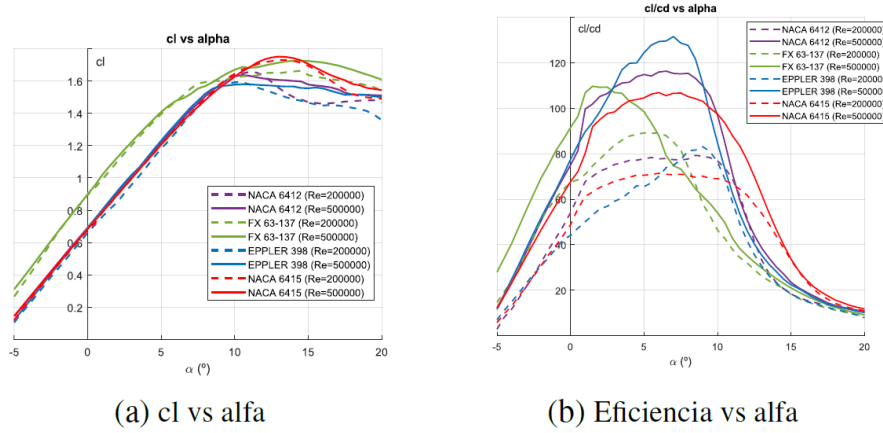


Figure 7: Comparative airfoil performance evaluated using XFLR5

| Airfoil    | $C_{L_{max}}$ | Drag Behaviour | Low-Re Suitability | Remarks        |
|------------|---------------|----------------|--------------------|----------------|
| NACA 6412  | High          | Moderate       | Good               | Higher drag    |
| NACA 6415  | Very High     | Moderate       | Very Good          | Strong lift    |
| Eppler 66  | Moderate-High | Low            | Excellent          | Efficient      |
| Eppler 210 | High          | Moderate       | Good               | Balanced       |
| Eppler 398 | High          | Moderate-High  | Good               | Less efficient |

Table 3: Airfoil comparison

The results indicated that cambered airfoils such as NACA 6415 provide higher lift generation, making them suitable for the wing root, while Eppler 66 offers superior efficiency at low Reynolds numbers and is therefore selected for the wing tip. The symmetric NACA 0012 airfoil is selected for the tail to ensure balanced control response.

## 4.2 Wing Design

Following the airfoil selection, the wing geometry was developed through parametric studies in XFLR5, considering aerodynamic efficiency, structural feasibility, and competition constraints.

The wing adopts a high-wing configuration to enhance inherent stability and simplify payload integration. A tapered planform was selected to improve lift distribution and reduce induced drag.

| Parameter    | Value              |
|--------------|--------------------|
| Wingspan     | 2.6 m              |
| Wing Area    | 0.5 m <sup>2</sup> |
| Aspect Ratio | 8.16               |
| Root Chord   | 0.20 m             |
| Tip Chord    | 0.164 m            |
| Taper Ratio  | 0.82               |
| Sweep Angle  | 12°                |

Table 4: Wing geometry parameters

The relatively high aspect ratio reduces induced drag, while the taper ratio improves lift

distribution across the span. The low wing loading ensures reduced stall speed and improved takeoff performance.

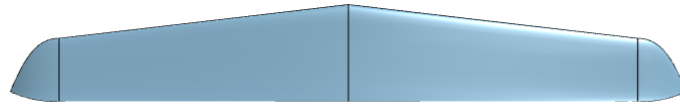


Figure 8: Wing configuration

### 4.3 Tail Design

The empennage was designed to provide adequate longitudinal and directional stability while minimizing aerodynamic drag and structural weight. Multiple tail configurations were initially considered, including conventional tail, T-tail, and V-tail configurations. Each configuration was evaluated based on aerodynamic efficiency, structural complexity, weight, and control effectiveness.

The conventional tail offers strong control authority but results in higher drag due to increased wetted area. The T-tail configuration improves aerodynamic cleanliness but introduces structural complexity and potential deep stall issues. In contrast, the V-tail configuration combines the functions of both horizontal and vertical stabilizers into two inclined surfaces, reducing overall surface area and associated drag.

Based on this trade-off analysis, the V-tail configuration was selected as the most suitable option for the aircraft, as it provides a good balance between aerodynamic efficiency and structural simplicity.

| Configuration     | Drag     | Weight   | Complexity |
|-------------------|----------|----------|------------|
| Conventional Tail | High     | Moderate | Low        |
| T-Tail            | Moderate | High     | High       |
| V-Tail            | Low      | Low      | Moderate   |

Table 5: Tail configuration comparison

The final tail geometry is summarized below:

| Parameter     | Value               |
|---------------|---------------------|
| Configuration | V-tail              |
| Tail Arm      | 0.48 m              |
| Tail Area     | 0.06 m <sup>2</sup> |
| V-angle       | 30°–40°             |
| Tail Airfoil  | NACA 0012           |

Table 6: Tail geometry parameters

The symmetric NACA 0012 airfoil was selected for the tail to ensure consistent aerodynamic

behavior for both positive and negative deflections. This is particularly important for the V-tail configuration, where the same control surfaces (ruddervators) are responsible for both pitch and yaw control.

The tail arm provides sufficient moment arm to generate stabilizing forces, while the overall tail sizing ensures adequate stability across the flight envelope. Although the V-tail configuration slightly reduces control effectiveness compared to a conventional tail, this is compensated through appropriate control surface sizing and deflection.

Overall, the selected tail configuration achieves reduced drag and weight while maintaining sufficient stability and control authority for the mission requirements.

## 5 Stability Analysis

The stability characteristics of the aircraft were evaluated through numerical simulations performed in XFLR5. Both static and dynamic stability analyses were conducted to assess the aircraft's response to disturbances and to ensure safe and predictable flight behavior across the operational envelope.

### 5.1 Static Stability Analysis

The static stability of the aircraft was investigated by analyzing the variation of aerodynamic coefficients with respect to angle of attack. The lift curve ( $C_L$  vs  $\alpha$ ) exhibits a linear trend within the operating range, indicating predictable aerodynamic behavior and confirming that the aircraft operates within the linear aerodynamic region.

The drag polar ( $C_L$  vs  $C_D$ ) follows a smooth quadratic relationship, demonstrating efficient aerodynamic performance without abnormal drag rise. This behavior is consistent with a well-designed wing configuration.

The pitching moment coefficient ( $C_M$  vs  $\alpha$ ) shows a negative slope, satisfying the condition for longitudinal static stability:

$$C_{M_\alpha} < 0$$

This indicates that any increase in angle of attack produces a restoring moment, allowing the aircraft to return to equilibrium after a disturbance. The trim condition is achieved at a small positive angle of attack, which lies within the desired operating range.

The key static stability parameters are summarized in Table 7.

| Parameter                             | Value                       | Observation            |
|---------------------------------------|-----------------------------|------------------------|
| Trim Angle ( $\alpha_{C_M=0}$ )       | $\approx 1.5^\circ$         | Within operating range |
| Lift Curve Slope ( $C_{L_\alpha}$ )   | $\approx 0.10/\text{deg}$   | Linear behavior        |
| Pitch Moment Slope ( $C_{M_\alpha}$ ) | $\approx -0.012/\text{deg}$ | Stable                 |
| Maximum Lift Coefficient              | $\approx 1.3$               | Adequate for payload   |
| Maximum $C_L/C_D$                     | $\approx 18\text{--}20$     | Efficient cruise       |

Table 7: Summary of static stability parameters

Overall, the static stability results confirm that the aircraft exhibits stable aerodynamic behavior with sufficient restoring characteristics in pitch.

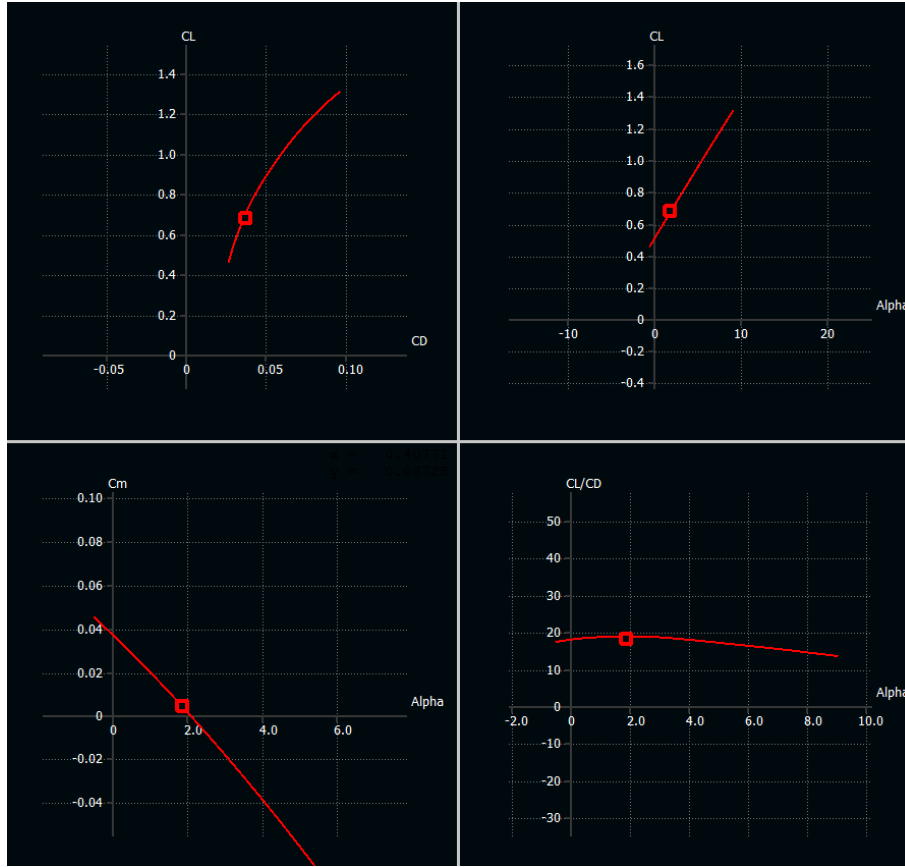


Figure 9: Static stability characteristics: variation of lift coefficient ( $C_L$ ) with drag coefficient ( $C_D$ ), lift coefficient ( $C_L$ ) with angle of attack ( $\alpha$ ), pitching moment coefficient ( $C_m$ ) with angle of attack ( $\alpha$ ), and lift-to-drag ratio ( $C_L/C_D$ ) with angle of attack ( $\alpha$ ).

## 5.2 Dynamic Stability Analysis

The dynamic stability of the aircraft was evaluated by analyzing the time-domain response of key state variables following an initial disturbance. The response was studied for both longitudinal and lateral-directional modes.

### Longitudinal Dynamic Response

The longitudinal response is characterized by the variation of forward velocity ( $u$ ), vertical velocity ( $w$ ), pitch rate ( $q$ ), and pitch angle ( $\theta$ ). The forward velocity exhibits small deviations that decay over time, indicating stable behavior.

The vertical velocity and pitch rate show rapid initial responses followed by damped oscillations, corresponding to the short-period mode. The fast decay of these oscillations indicates that the short-period mode is well damped.

The pitch angle demonstrates a slower oscillatory response with gradual decay, which is characteristic of the phugoid mode. The damping of this oscillation confirms that the phugoid mode is stable.

### Lateral-Directional Dynamic Response

The lateral-directional response is analyzed through side velocity ( $v$ ), roll rate ( $p$ ), yaw rate ( $r$ ), and roll angle ( $\phi$ ). The side velocity stabilizes quickly after a small disturbance, indicating good

lateral damping.

The roll and yaw rates exhibit oscillatory behavior with decreasing amplitude, which corresponds to a stable Dutch roll mode. The roll angle returns to equilibrium without divergence, confirming that the spiral mode is stable.

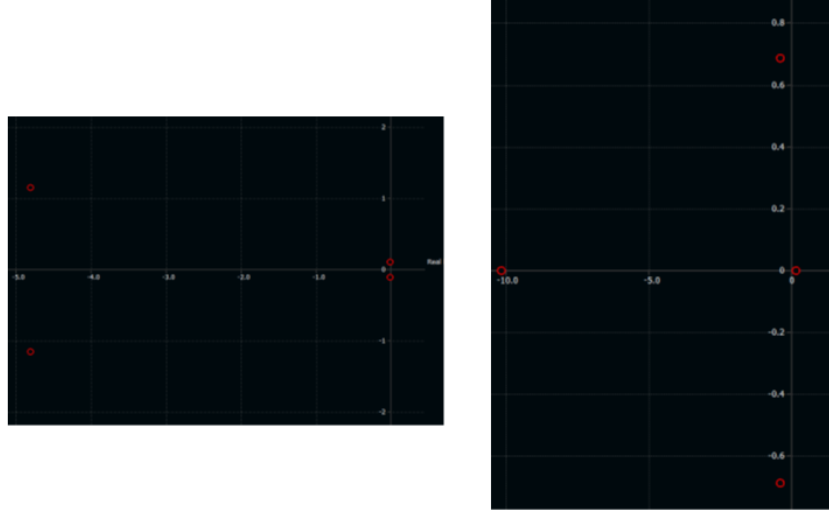


Figure 10: Lateral (left) and Longitudinal (right) root locus point

### 5.3 Control Surface Design

The control surfaces were designed to ensure adequate maneuverability and control authority while maintaining aerodynamic efficiency. The sizing of the control surfaces was based on standard empirical ratios for small-scale aircraft and validated through XFLR5 simulations.

#### Aileron Design

The ailerons are responsible for roll control and were sized as a fraction of the wing semi-span and chord. The spanwise extent of the aileron was selected as approximately 30% of the semi-span, while the chord was chosen as 25% of the local wing chord to provide sufficient rolling moment without inducing excessive drag.

The aileron dimensions are calculated as follows:

$$b_a = 0.3 \times \frac{b}{2} = 0.3 \times 1.3 = 0.39 \text{ m} \quad (10)$$

$$c_a = 0.25 \times c = 0.25 \times 0.20 = 0.05 \text{ m} \quad (11)$$

The selected dimensions provide sufficient roll authority while maintaining smooth aerodynamic performance.

#### V-Tail Control Surfaces (Ruddervators)

The V-tail incorporates ruddervators, which combine the functions of both elevator and rudder. Due to the reduced effectiveness of V-tail configurations compared to conventional tails, a larger control surface fraction was selected.

| Parameter        | Value                     |
|------------------|---------------------------|
| Aileron Span     | 0.39 m                    |
| Aileron Chord    | 0.05 m                    |
| Span Fraction    | 30% semi-span             |
| Chord Fraction   | 25% chord                 |
| Deflection Range | $\pm 15^\circ - 20^\circ$ |

Table 8: Aileron design parameters

The control surface chord was defined as approximately 35% of the tail chord:

$$c_{rv} = 0.35 \times c_{tail} = 0.35 \times 0.10 = 0.035 \text{ m} \quad (12)$$

| Parameter             | Value                     |
|-----------------------|---------------------------|
| Tail Span (one side)  | 0.64 m                    |
| Tail Area             | 0.06 m <sup>2</sup>       |
| Control Surface Chord | 0.035 m                   |
| Chord Fraction        | 35% tail chord            |
| Elevator Deflection   | $\pm 15^\circ - 20^\circ$ |
| Rudder Deflection     | $\pm 20^\circ - 25^\circ$ |

Table 9: V-tail control surface parameters

The increased control surface size ensures sufficient pitch and yaw authority despite the geometric limitations of the V-tail configuration.

### Control Effectiveness

The control surfaces were sized to satisfy the following requirements:

- Adequate rolling moment for maneuverability
- Sufficient pitch control for trim and stability
- Effective yaw control despite V-tail configuration

The final configuration was validated using XFLR5 to confirm that the control surfaces provide sufficient authority without causing excessive drag or instability.

### 5.4 Control Summary

The control surface design achieves a balance between effectiveness and efficiency. The selected sizing ensures stable and responsive aircraft behavior across all flight conditions, while maintaining aerodynamic performance suitable for both payload and endurance missions.

## 6 Structural Design and Analysis

The structural design of the aircraft was developed with the objective of achieving a high strength-to-weight ratio while ensuring reliability under expected flight loads. The design integrates lightweight composite materials with simple manufacturing techniques to maintain structural efficiency and ease of assembly.

## 6.1 Material Properties

The primary materials used in the structural design are summarized in Table 10.

Table 10: Material Properties

| Material            | Young's Modulus (GPa)     | Density (kg/m <sup>3</sup> ) | Poisson's Ratio |
|---------------------|---------------------------|------------------------------|-----------------|
| Carbon Fiber (CFRP) | $E_1 = 135$<br>$E_2 = 10$ | 1550                         | 0.30            |
| Balsa Wood          | 0.5                       | 160                          | 0.30            |
| ABS Plastic         | 2.1                       | 1050                         | 0.35            |

## 6.2 Design Approach

The structural configuration follows a approach, where the external skin contributes to load bearing while internal elements provide reinforcement and shape stability. The main objectives of the design are:

- Minimizing structural weight
- Ensuring sufficient stiffness to avoid excessive deformation
- Maintaining robustness under payload and landing loads

The structure is designed to withstand aerodynamic loads during cruise, maneuver loads, and impact loads during landing.

## 6.3 Wing Structure

The wing is designed as a lightweight composite structure combining carbon fiber and balsa wood.

- Span: 1.3 m per wing (2.6 m total)
- Outer skin: Carbon fiber composite
- Internal structure: Balsa wood ribs

The carbon fiber skin provides high tensile strength and stiffness, allowing the wing to resist bending moments generated during flight. The internal balsa ribs maintain the aerodynamic profile and distribute loads across the span.

This hybrid structure provides a high stiffness-to-weight ratio while remaining easy to manufacture.

The wing adopts a high-wing configuration, which improves stability and enhances load distribution. The attachment to the fuselage is achieved using mechanical fasteners, ensuring a rigid connection capable of transferring aerodynamic loads while allowing easy assembly and maintenance.

## 6.4 Fuselage Structure

### 6.4.1 Internal Layout

The internal arrangement follows a longitudinal distribution:

1. Front section: Motor
2. Central section: Payload compartment
3. Rear section: Electronics and control systems

The payload consists of two water packs and one rice pack within a box of dimensions  $16 \times 12 \times 8.5$  cm. The payload is positioned near the center of gravity to ensure stability during flight. Any unused internal volume within the payload bay is filled with hard foam, which prevents component movement, absorbs vibrations, and contributes to structural damping.

#### 6.4.2 Structural Reinforcement

A wooden exoskeleton is used as the primary internal load-bearing structure. It provides rigidity against bending and supports the payload loads. This ensures enough reinforcement while maintaining the light weight.

#### 6.5 Tail Structure

The aircraft employs a V-tail configuration, which reduces structural weight and aerodynamic drag compared to conventional tail designs.

The tail is attached to the fuselage with reinforced connections to withstand aerodynamic and vibrational loads. These reinforcements ensure adequate stiffness and prevent structural instabilities such as flutter.

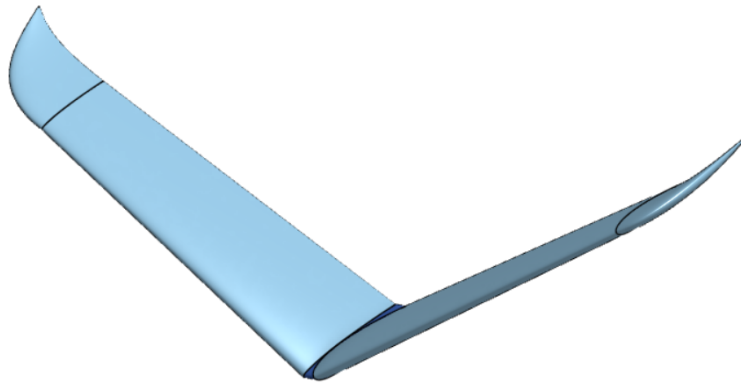


Figure 11: V-tail configuration of the aircraft showing the aerodynamic shape

#### 6.6 Telemetry Box Integration

The telemetry box is mounted on top of the fuselage behind the wing. This location minimizes interference with internal components and maintains aerodynamic balance.

Local structural reinforcement is provided to support the telemetry system without compromising fuselage integrity.

#### 6.7 Structural Performance Considerations

The structure is designed considering the primary load cases encountered during operation:

- Bending loads on the wings due to lift

- Shear loads at the wing-fuselage junction
- Impact loads during landing
- Dynamic loads from payload and onboard systems

The use of carbon fiber improves resistance to bending, while the internal wooden structure ensures effective load distribution and manufacturability.

## 6.8 Summary

The final structural design achieves a balance between lightweight construction and mechanical strength. The combination of carbon fiber skin, balsa wood ribs, and reinforced fuselage ensures that the aircraft can safely withstand operational loads while maintaining high efficiency.

The modular configuration also allows for easy assembly, maintenance, and future modifications.

# 7 Aircraft Performance

This section presents the performance calculations for the aircraft at a maximum take-off weight of 3.0 kg, using the baseline parameters provided

## 7.1 Stall Speed

The stall speed is determined from the lift equation at the point where lift equals weight:

$$V_s = \sqrt{\frac{2W}{\rho S C_{L_{\max}}}} \quad (13)$$

$$V_s = 8.59 \text{ m/s} \quad (14)$$

## 7.2 Cruise Speed

The cruise speed is determined assuming steady, level flight where lift equals weight:

$$V = \sqrt{\frac{2W}{\rho S C_L}} \quad (15)$$

$$V = 12.65 \text{ m/s} \quad (16)$$

## 7.3 Maximum Speed

The maximum speed is estimated using the power balance condition, where the available power equals the required power. Assuming that power required varies with the cube of velocity:

$$V_{\max} = V \left( \frac{P_{\text{avail}}}{P_{\text{req}}} \right)^{1/3} \quad (17)$$

$$V_{\max} = 16.45 \text{ m/s} \quad (18)$$

#### 7.4 Thrust-to-Weight Ratio

The thrust-to-weight ratio is defined as the ratio of the available thrust to the aircraft weight:

$$\frac{T}{W} = 3 \quad (19)$$

$$\frac{T}{W} = 0.88 \quad (20)$$

#### 7.5 Rate of Climb

The rate of climb is determined from the excess power available:

$$ROC = \frac{P_{avail} - P_{req}}{W} \quad (21)$$

$$ROC = 10.20 \text{ m/s} \quad (22)$$

#### 7.6 Takeoff Speed

The takeoff speed is approximated as a multiple of the stall speed:

$$V_{TO} = 1.2 V_s \quad (23)$$

$$V_{TO} = 10.31 \text{ m/s} \quad (24)$$

#### 7.7 Endurance

The endurance is calculated from the ratio of available energy to power consumption:

$$t = \frac{E}{P_{req}} \quad (25)$$

$$t = 17.8 \text{ min} \quad (26)$$

#### 7.8 Takeoff Distance

The takeoff distance is estimated assuming constant acceleration.

First, the acceleration is determined from:

$$a = \frac{T}{m} \quad (27)$$

$$a = 8.63 \text{ m/s}^2 \quad (28)$$

The takeoff distance is then calculated using:

$$S_{TO} = \frac{V_{TO}^2}{2a} \quad (29)$$

$$S_{TO} = 6.156 \text{ m} \quad (30)$$

## 7.9 Range

The range is calculated as the product of cruise speed and endurance:

$$Range = V \times t \quad (31)$$

$$Range = 13485 \text{ m} = 13.5 \text{ km} \quad (32)$$

## 7.10 Performance Summary

| Parameter                     | Value             |
|-------------------------------|-------------------|
| Stall Speed, $V_s$            | 8.59 m/s          |
| Cruise Speed, $V$             | 12.65 m/s         |
| Maximum Speed, $V_{\max}$     | 16.45 m/s         |
| Thrust-to-Weight Ratio, $T/W$ | 0.88              |
| Rate of Climb, $ROC$          | 10.20 m/s         |
| Takeoff Speed, $V_{TO}$       | 10.31 m/s         |
| Takeoff Distance, $S_{TO}$    | 6.2 m             |
| Endurance, $t$                | 17.8 min          |
| Range                         | 13.5 km (13485 m) |

Table 11: Performance Summary

# 8 Materials

The aircraft structure uses a hybrid material strategy. CFRP is used in primary load paths, while lightweight wood is used for ribs, internal frames, and secondary supports.

## 8.1 Material Selection Strategy

Material selection was performed using three complementary methods:

1. Ashby chart screening;
2. performance index method;
3. multi-criteria decision method.

The objective was to select materials that provide high stiffness, low density, adequate strength, and practical manufacturability.

## 8.2 Selected Aircraft Materials

Table 12: Selected aircraft materials

| Aircraft Part    | Material                       | Function                                     |
|------------------|--------------------------------|----------------------------------------------|
| Wing skins       | Plain-weave CFRP               | Aerodynamic surface and torsional stiffness  |
| Wing spars       | Unidirectional CFRP rods/tubes | Main bending load path                       |
| Wing ribs        | Balsa or lightweight plywood   | Maintains airfoil geometry                   |
| Fuselage shell   | CFRP or GFRP                   | Lightweight external structure               |
| Fuselage frames  | Plywood or balsa               | Internal support for payload and electronics |
| Tail boom        | Carbon fibre tube              | Connects fuselage and empennage              |
| Tail surfaces    | CFRP/balsa hybrid              | Lightweight stability surfaces               |
| Control surfaces | Balsa/CFRP sandwich            | Lightweight and stiff control surfaces       |
| Adhesive joints  | Epoxy resin                    | Structural bonding                           |
| Moulds           | 3D printed PLA                 | Low-cost composite manufacturing tooling     |

## 8.3 Ashby Chart Screening

For a lightweight UAV wing, the main requirement is a high stiffness-to-weight ratio. Therefore, Young's modulus and density were selected as the main screening properties.

The screening constraints were:

$$E \geq 30 \text{ GPa} \quad (33)$$

$$\rho \leq 3000 \text{ kg/m}^3 \quad (34)$$

The remaining candidate families were:

- Carbon Fibre Reinforced Polymer;
- Glass Fibre Reinforced Polymer;
- Aluminium alloy.

CFRP was located in the most favourable region due to its high specific stiffness.

## 8.4 Performance Index Method

Since the wing is bending-dominated and mass-critical, the material performance index is:

$$PI = \frac{\sqrt{E}}{\rho} \quad (35)$$

Table 13: Performance index comparison

| Material          | $E$ (GPa) | $\rho$ (kg/m <sup>3</sup> ) | PI      |
|-------------------|-----------|-----------------------------|---------|
| CFRP              | 70        | 1600                        | 0.00523 |
| Aluminium 6066-T6 | 69        | 2700                        | 0.00308 |
| GFRP              | 30        | 1900                        | 0.00288 |

CFRP achieves the highest performance index and is therefore the best material for minimizing structural mass while satisfying stiffness requirements.

### 8.5 Multi-Criteria Decision Method

A multi-criteria decision method was used to account for mechanical and practical design considerations.

The criteria were:

- Young's modulus;
- density;
- shear modulus;
- strength;
- manufacturability;
- cost.

Table 14: MCDM criteria weights

| Criterion         | Weight |
|-------------------|--------|
| Young's modulus   | 0.30   |
| Density           | 0.20   |
| Shear modulus     | 0.15   |
| Strength          | 0.15   |
| Manufacturability | 0.10   |
| Cost              | 0.10   |

Table 15: Final MCDM ranking

| Material  | Total Score |
|-----------|-------------|
| CFRP      | 0.876       |
| Aluminium | 0.794       |
| GFRP      | 0.591       |
| PLA       | 0.482       |

The MCDM results confirm that CFRP is the most suitable material for the main wing structure.

## 9 Manufacturing Plan

### 9.1 Skin Fabrication

The skins were manufactured using a manual wet layup combined with vacuum bagging. This method was selected due to its low cost, accessibility, and suitability for composite prototyping.

### 9.2 Materials

The composite system consisted of dry carbon fibre fabric and a two-part epoxy resin system. The fibre was supplied in dry form, requiring manual impregnation. The epoxy system was prepared using a mixing ratio of approximately 100:27 (resin:hardener by mass), ensuring proper curing conditions.

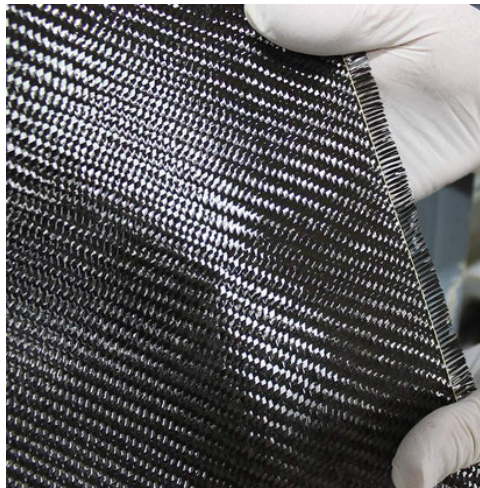


Figure 12: Carbon Fiber sheet

### 9.3 Mould Preparation

The moulds were fabricated using the CAD geometry of the aircraft and produced via 3D printing. Separate moulds were generated for the upper and lower wing surfaces.

After printing, the moulds were manually sanded to improve surface finish and dimensional accuracy. Due to limited tooling, sanding was performed using abrasive pads, which, although not optimal, provided a sufficient reduction in surface roughness to improve the final laminate quality.

### 9.4 Layup Process

Prior to layup, a release agent was applied to the mould surfaces to facilitate demoulding. In the absence of a dedicated release agent, a spray oil was used as an alternative, although this is not recommended for high-quality composite manufacturing.

The layup process followed a sequential manual impregnation approach:

- Initial application of a resin layer onto the mould surface
- Placement of the first carbon fibre ply
- Application of a second resin layer
- Placement of the second carbon fibre ply

- Final resin layer to ensure full impregnation

Each mould contained two plies of carbon fibre and three layers of resin.

The fibre orientation was selected at  $\pm 45^\circ$  relative to the wing span. This configuration provides enhanced in-plane shear stiffness and torsional resistance, which are critical for wing structures subjected to combined bending and aerodynamic loading. Unlike unidirectional layups, the  $\pm 45^\circ$  arrangement ensures a more balanced mechanical response, improving structural stability and reducing the risk of delamination under multi-axial stresses.

## 9.5 Vacuum Bagging and Curing

To compact the laminate and remove excess resin, a vacuum bagging process was applied. A protective layer was placed over the final laminate to prevent adhesion to the vacuum bag. Initially, plastic film was considered; however, due to handling difficulties, paper was used instead.

This resulted in partial bonding between the laminate and the protective layer, highlighting a limitation of the process and an area for improvement in future iterations (use of peel ply is recommended).

The laminate was then sealed in a vacuum bag and left to cure for approximately 24 hours under atmospheric pressure.



Figure 13: Vacuum Bagging for 24h

## 9.6 Spar Fabrication

The spars were conceptually designed to be manufactured using unidirectional carbon fibre to maximise axial stiffness along the span.

# 10 Internal Structure

The internal ribs were fabricated using laser cutting from lightweight wood sheets available at the FabLab. Although balsa wood would have been the optimal material due to its superior density-to-stiffness ratio, an alternative wood was used due to availability constraints. Nevertheless, the selected material provided comparable lightweight characteristics.

The rib geometry was extracted from the wing CAD model and processed using the laser cutting software to ensure dimensional accuracy and proper alignment with the spars.



Figure 14: Laser cut airfoil ribs

### 10.1 Assembly

The internal structure is designed to be assembled by mounting the ribs onto the spars, forming a rigid internal framework. This structure is then integrated with the wing skins to create a closed wing box.

Although carbon fibre provides a high stiffness-to-weight ratio, the skin alone is insufficient to sustain structural loads efficiently. The internal rib-spar structure is therefore essential to:

- Maintain aerodynamic shape
- Increase bending stiffness
- Improve load distribution

This hybrid configuration results in a lightweight yet structurally efficient wing system.

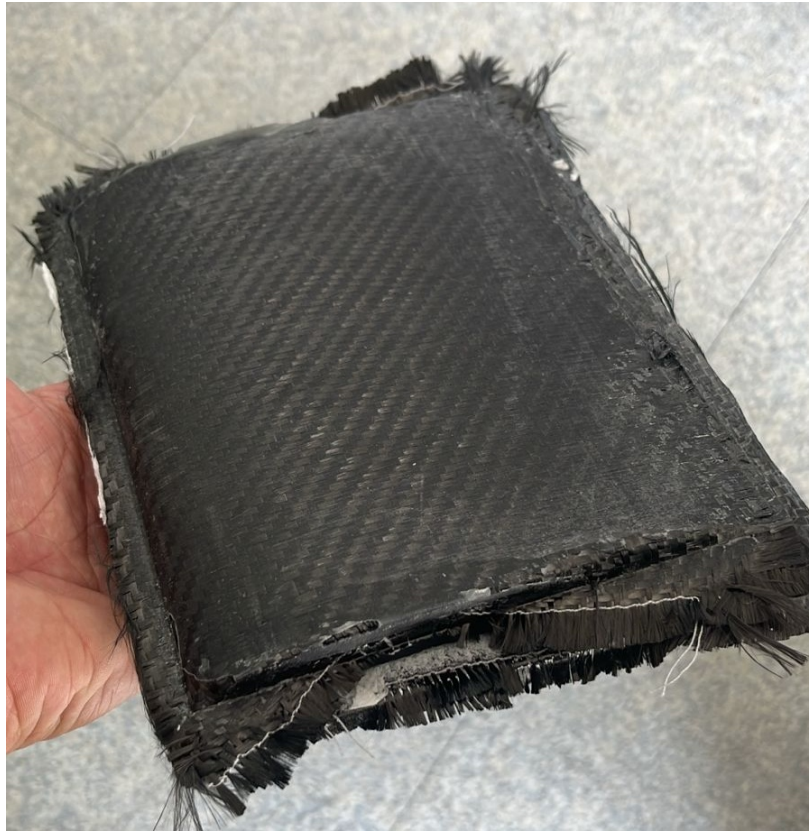


Figure 15: Final wing assembly without trimming

#### 10.1.1 Observations

Post-processing steps such as trimming and surface finishing are still in process and the intention to present the raw manufactured component.

## 11 Lessons learnt

The project highlighted the importance of effective time management, as delays in earlier stages limited the depth of work that could be carried out in structural design and manufacturing. Additionally, challenges in aerodynamic selection emphasized the need for a more iterative and well-planned design approach. These experiences underline the importance of better scheduling and balanced effort across all project phases in future work.

## 12 Conclusion

The final aircraft design meets the main objectives of the project, demonstrating stable flight characteristics, adequate payload capability, and efficient aerodynamic performance. The stability analysis confirms safe and predictable behavior, while the structural design ensures strength with low weight. Overall, the aircraft is a feasible and well-balanced solution for the intended mission.

## 13 Bibliography

1. M. H. Sadraey. *Aircraft Design – A Systems Engineering Approach*. John Wiley & Sons, 2013.
2. S. A. Brandt and R. J. Starnes. *Preliminary Aircraft Design and Performance Estimation*. University of North Dakota, 2023.
3. Emax. *GT2826/04 1090 KV Brushless Motor Datasheet*. Available at: <https://www.emaxus.com>.
4. APC Propellers. *Propeller Performance Database*. Available at: <https://www.apcprop.com>.
5. Hobbywing. *Skywalker 60A V2 Electronic Speed Controller Manual*.
6. eCalc AG. *eCalc – Propeller, Motor and Battery Calculator*. Available at: <https://www.ecalc.ch>.
7. M. F. Ashby. *Materials Selection in Mechanical Design*, 4th ed., Butterworth-Heinemann, 2011.
8. M. H. Sadraey. *Aircraft Design – A Systems Engineering Approach*. John Wiley & Sons, 2013.
9. S. A. Brandt and R. J. Starnes. *Preliminary Aircraft Design and Performance Estimation*. University of North Dakota, 2023.
10. P. S. Bernstein and J. L. Stewart. “Composite Materials for Aircraft Structures,” *AIAA Education Series*, 2016.

## Appendix - I Detailed cost breakdown

### 13.0.1 Electronics and Propulsion Budget

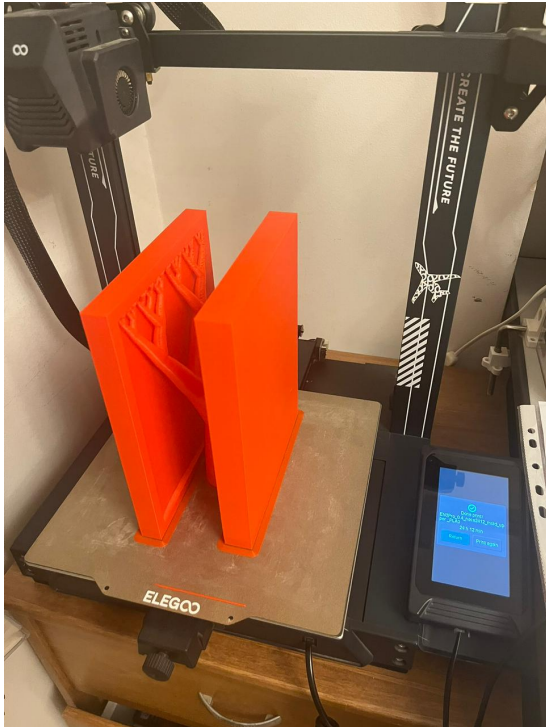
Table 16: Electronics and propulsion cost breakdown

| Component                            | Quantity | Estimated Cost [EUR] |
|--------------------------------------|----------|----------------------|
| Emax GT2826/04 1090KV motor          | 1        | 43                   |
| Tattu 2300mAh 4S 75C LiPo battery    | 1        | 36                   |
| Tattu 650mAh 2S HV secondary battery | 1        | 11                   |
| SKYWALKER 60A V2 ESC                 | 1        | 25                   |
| FS-i6 radio + receiver set           | 1        | 55                   |
| EMAX servos                          | 6        | 48                   |
| UBEC 5A                              | 1        | 8                    |
| 40A safety switch                    | 1        | 6                    |
| Wiring and connectors                | 1 set    | 15                   |
| <b>Total electronics cost</b>        | –        | <b>247</b>           |

### 13.0.2 Materials and Fabrication Budget

Table 17: Materials and fabrication cost breakdown

| Material / Item                           | Quantity    | Estimated Cost [EUR] |
|-------------------------------------------|-------------|----------------------|
| Carbon fibre fabric (critical areas only) | reduced set | 30                   |
| Epoxy resin and hardener                  | reduced set | 25                   |
| Carbon fibre tubes (spar and tail boom)   | 1–2 pieces  | 25                   |
| Balsa / plywood sheets (ribs and frames)  | 1 set       | 25                   |
| PLA filament (moulds)                     | partial use | 15                   |
| Vacuum bagging film and sealant tape      | reduced set | 18                   |
| Peel ply / release film                   | reduced set | 10                   |
| Release agent / wax                       | 1 set       | 8                    |
| <b>Total materials cost</b>               | –           | <b>156</b>           |



(a) 3D Printed Mold

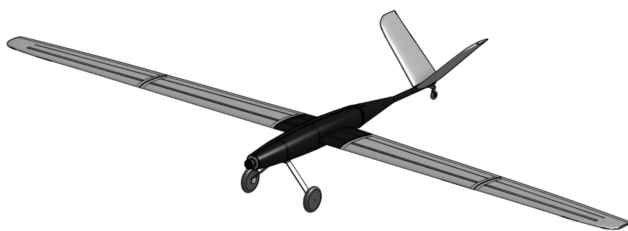


(b) Mixing Epoxy A and B

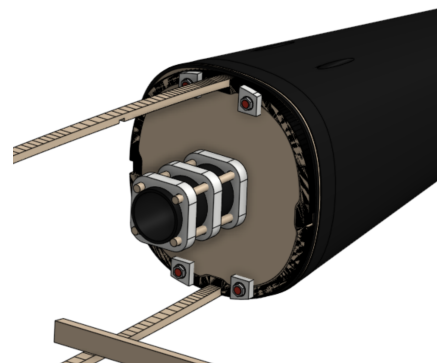


(c) Carbon Fiber Layup

Figure 16: Manufacturing process steps



(a) Plane CAD model



(b) Internal tube structure

Figure 17: CAD model

## Appendix - II Aircraft drawings

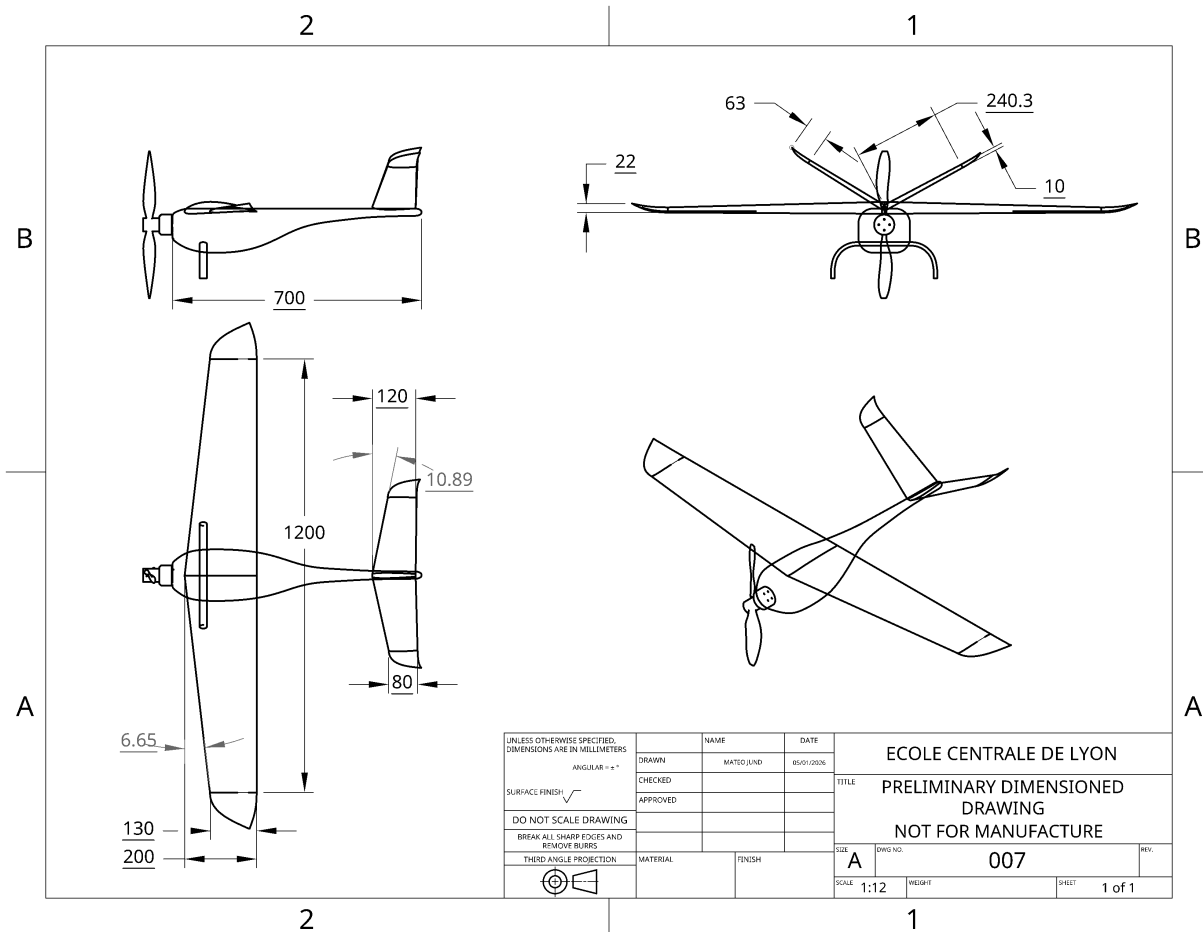


Figure 18: Aircraft schematic (with dimensions)